COLLEGE OF ARTS AND TECHNOLOGY

CIIT

Pharmacy Inventory : Metabase Dashboard and Report

A Metabase-Integrated Web Reporting System for Pharmacy Inventory, Sales, and Supplier Management

Ciara Frances Comendador
Mhikaela Galo
Amanda Grace Mangubat

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INTRODUCTION

1.1 System Overview

The EMC001 Dashboard is a web-based analytics platform developed for real-time monitoring and reporting on medicine inventory, sales, and supplier efficiency in a pharmacy setting. It integrates with Metabase to generate interactive dashboards and reports, all embedded into a custom website with modern UI features such as a collapsible sidebar, dark/light mode toggle, and responsive layout.

1.2 Dataset Explanation

The dataset used in this system simulates pharmacy operations and contains structured information across key domains:

1. **Medicines:** Details of each medicine including name, category, brand, and unit price.
2. **Inventory:** Current stock levels, reorder thresholds, expiration dates, and batch tracking for each medicine.
3. **Transactions:** Sales transaction logs that record the quantity sold, date of purchase, total price, and customer details.
4. **Suppliers:** Information about pharmaceutical suppliers, including supplier names, delivery records, and contact information.
5. **Users (Admins):** Login information and permissions for users managing the dashboard.

This dataset allows for insights such as:

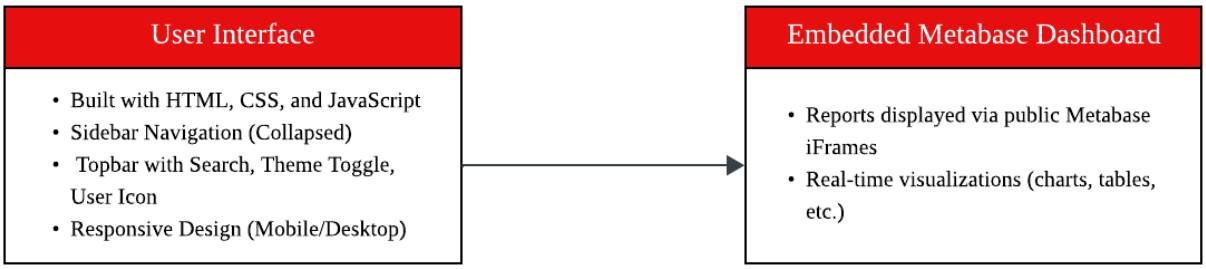
- Identifying low stock medicines and those nearing expiration
- Visualizing monthly and daily revenue trends
- Tracking top-performing products and supplier reliability
- Monitoring inventory levels over time to avoid stockouts

SYSTEM ARCHITECTURE

The system architecture is composed of the following components:

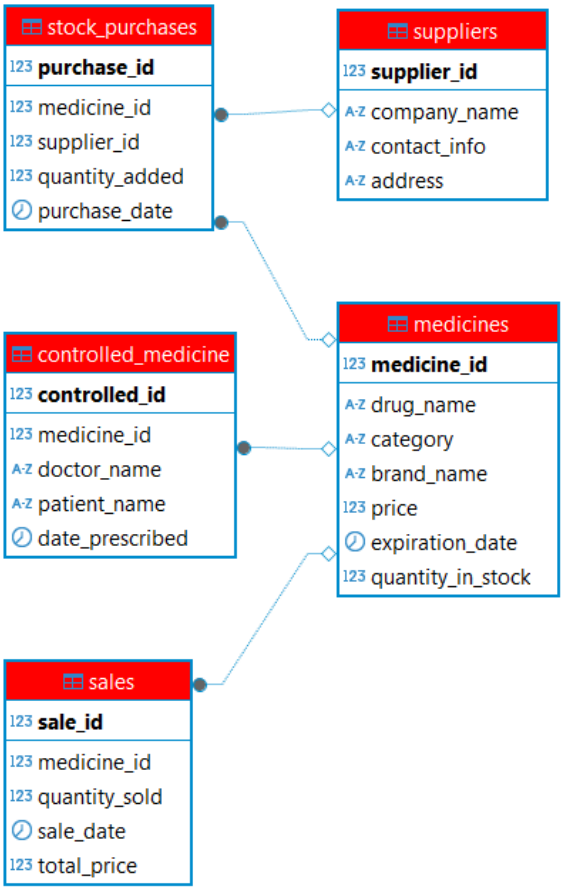
Frontend

Built with HTML, CSS, and JavaScript. Custom layouts include a collapsible sidebar, light/dark mode toggle, and embedded Metabase iFrames.



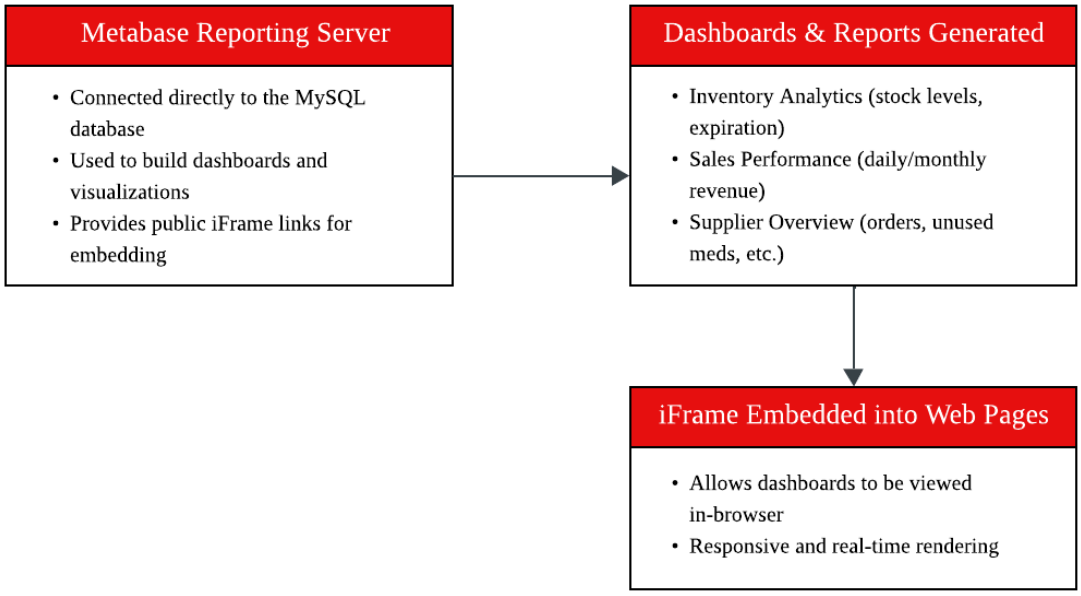
Backend

MySQL database storing pharmacy inventory and transaction data.



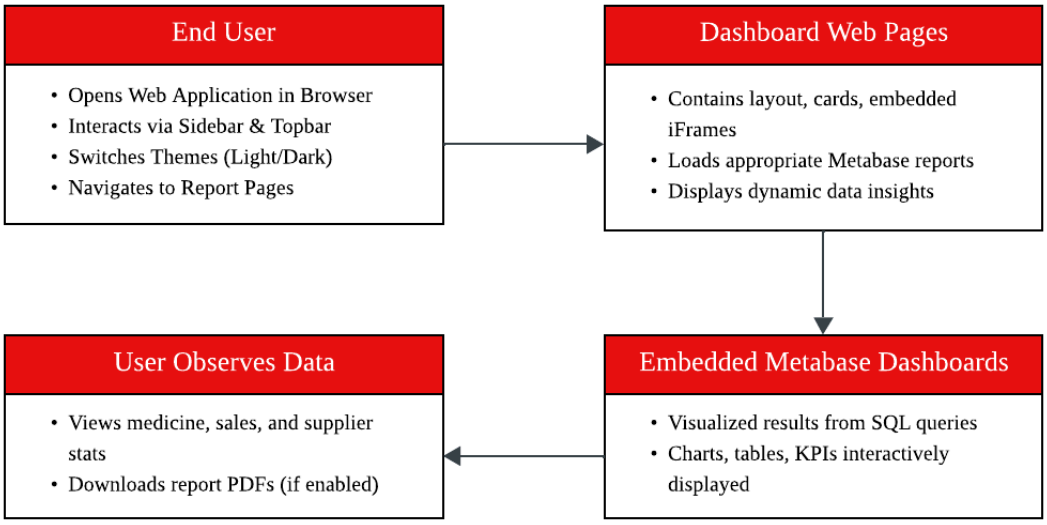
Analytics Layer

Metabase used for creating dashboards and reports.



User Interaction

Users interact via the web interface, which dynamically loads reports.

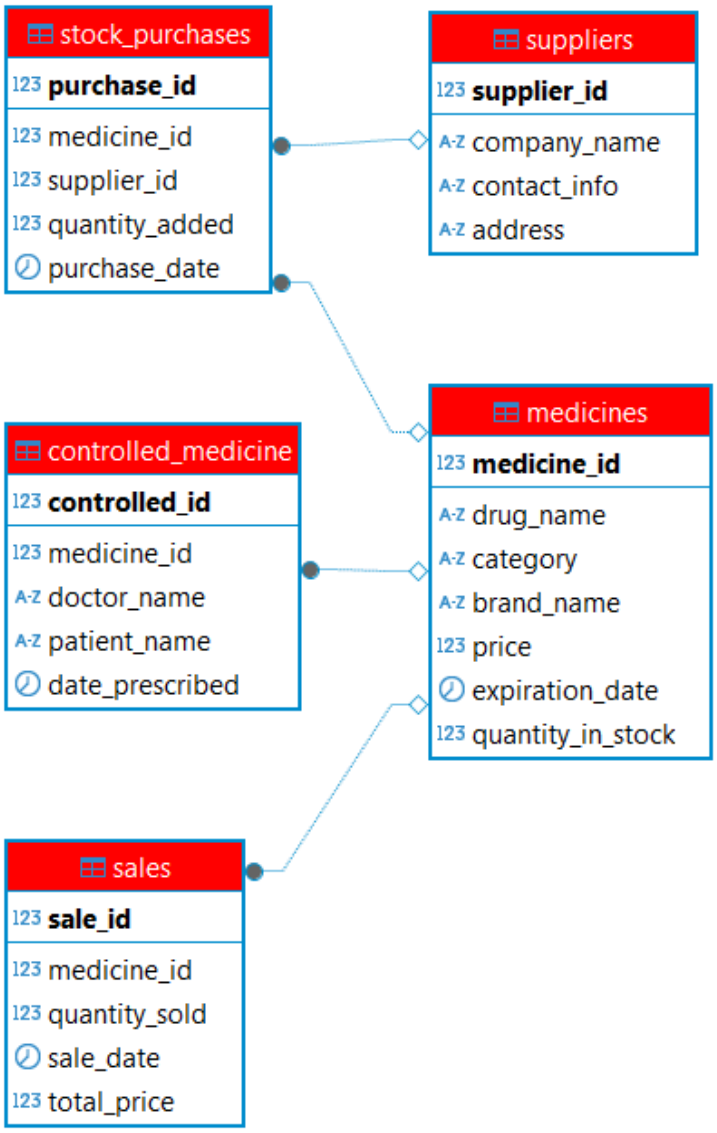


DATABASE SCHEMA

The database, named PharmacyInventory, includes key tables such as:

1. Medicines
2. Inventory
3. Transactions
4. Suppliers
5. Users

These tables are interrelated to allow report generation on stock levels, sales, supplier performance, and expiration tracking.



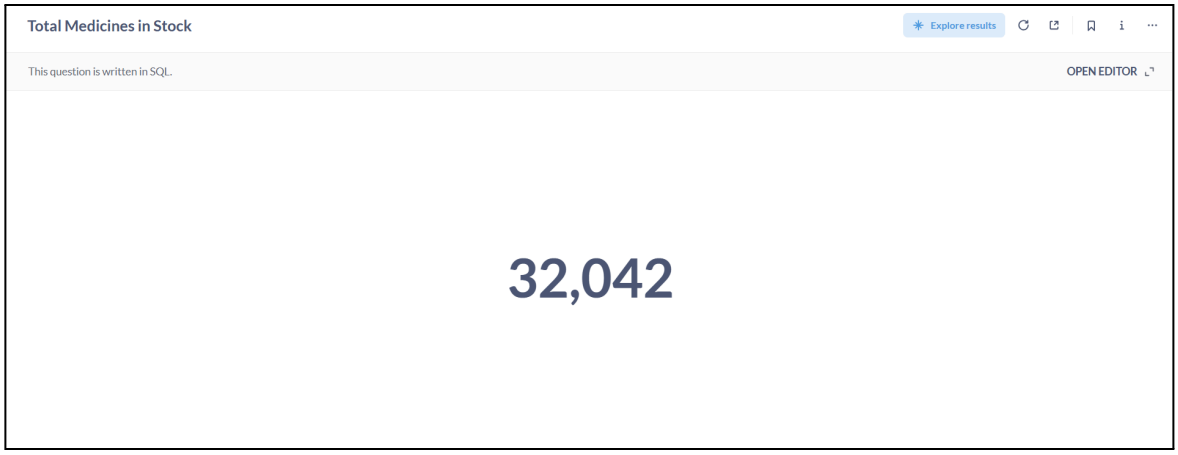
METABASE DASHBOARDS & REPORTS

The system integrates Metabase visualizations into the web platform using iFrames. These are categorized into three sections:

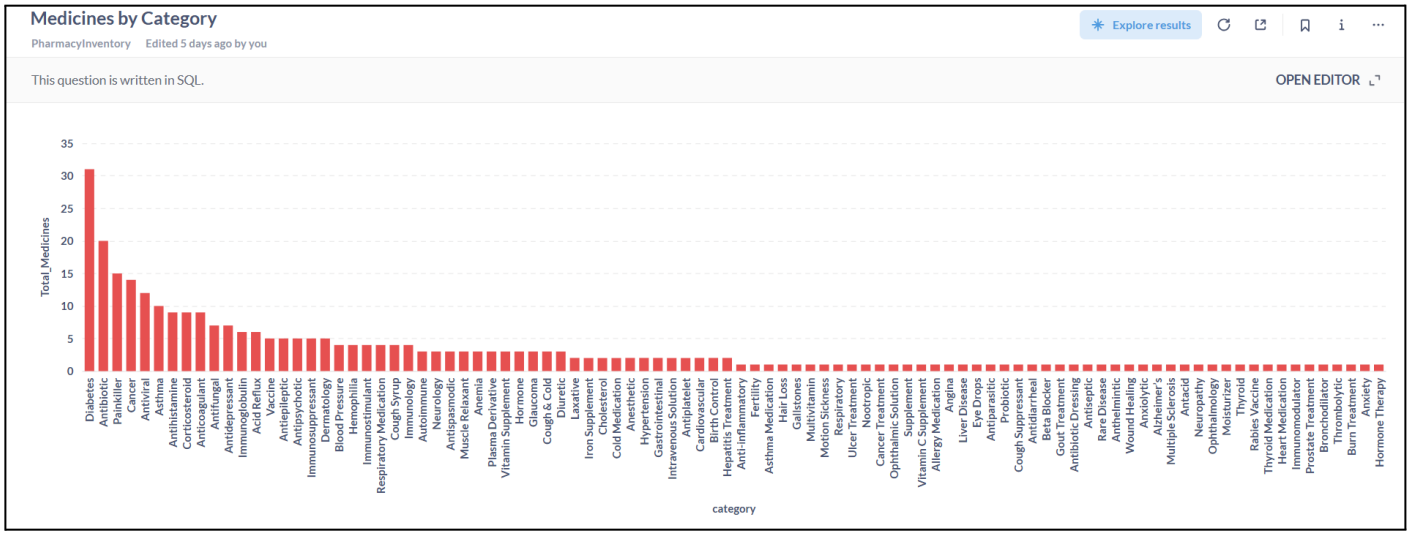
Dropdown #1: Medicine & Inventory Analytics

Medicine Insights Dashboard:

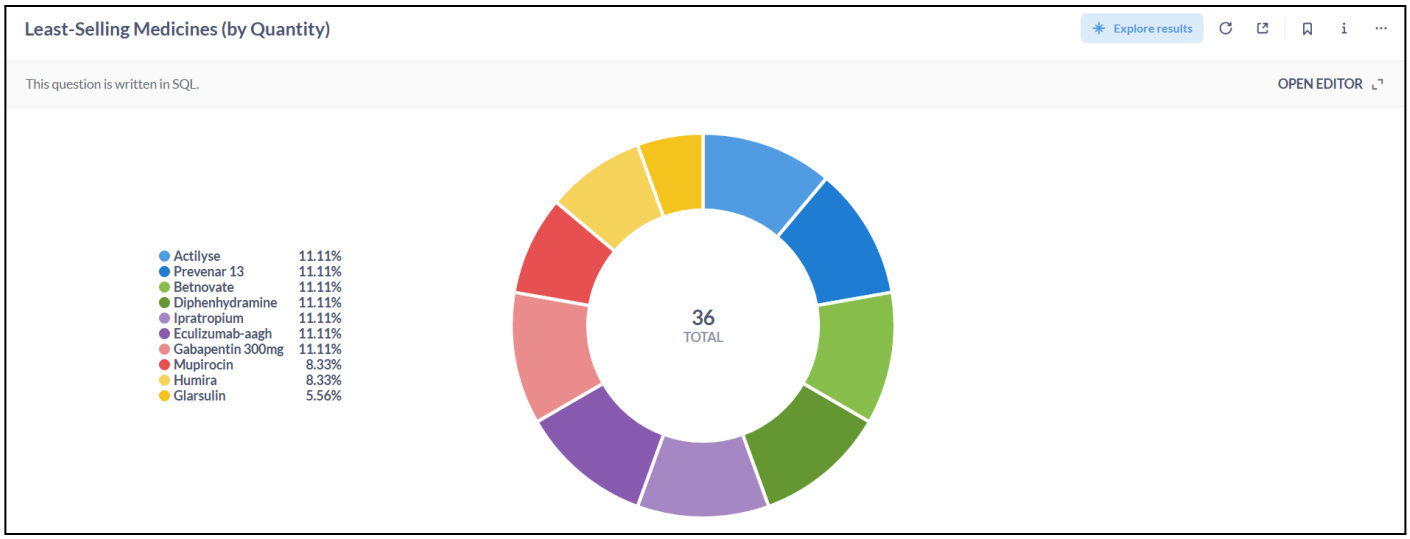
Total Medicines In Stock



Medicines by Category



Least-Selling Medicines (by Quantity)



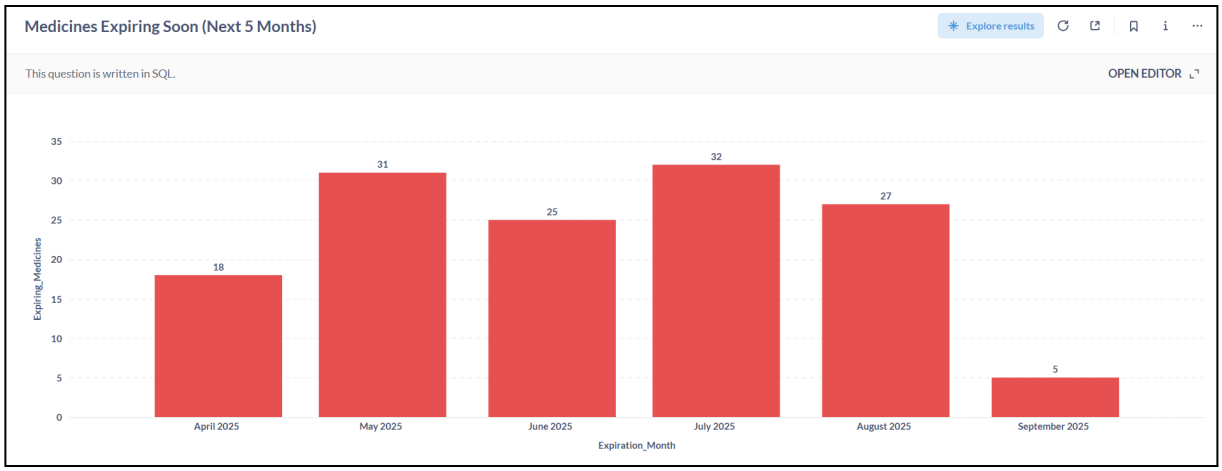
Low Stock Medicines (≤ 20 units)

Low Stock Medicines (≤ 20 units)

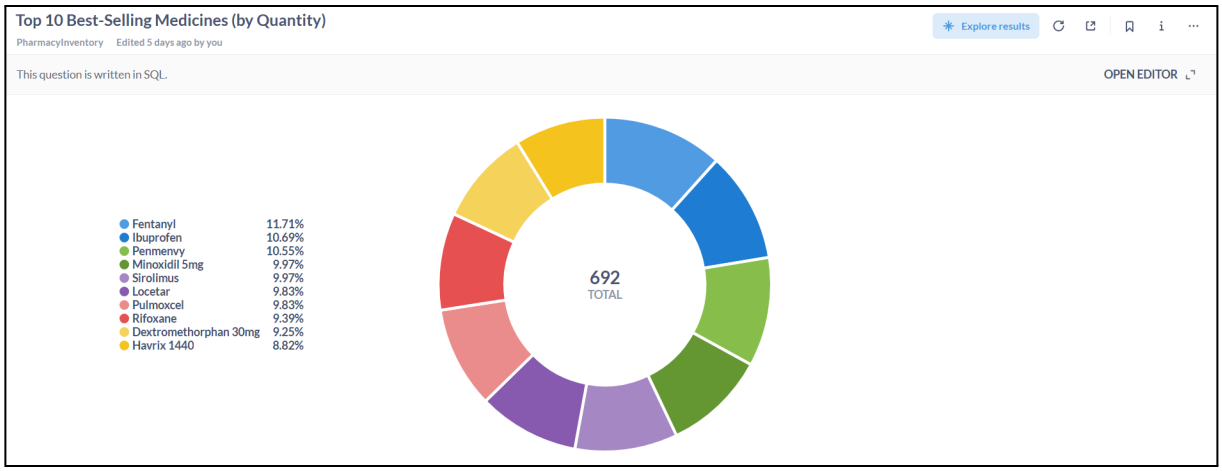
This question is written in SQL.

drug_name	category	brand_name...	quantity_in_sto...
Ferlin	Iron Supplement	HemoIron	10
Trulicity	Diabetes	GlucoTune	10
Penmenvy	Antiviral	MediCure	10
Morphine	Painkiller	MS Contin	12
Neo-Bromexan For...	Cold Medication	FluBlock	12
Resmetirom	Liver Disease	HepatoCare	12
Octanate	Hemophilia	FactorX	12
Xultophy	Diabetes	GlucoBlend	13
Trastuzumab	Cancer	OncoShield	13
Glarsulin	Diabetes	InsuFlow	14
Benadryl	Antihistamine	HistaRelief	14
Albuterol	Asthma	BreatheWell	14
Lexibase	Antacid	AcidGuard	14
Dramelin	Motion Sickness	TravelEase	14
Clotrimazole	Antifungal	SkinGuard	15
Nivolumab	Cancer	TumorBlock	16
Tergecef	Antibiotic	BactoCef	16

Medicines Expiring Soon (Next 5 Months)



Top 10 Best-Selling Medicines (by Quantity)



Inventory Status Report

Inventory Status Report

This question is written in SQL.

drug_name	category	brand_name	quantity_in_stock
Paracetamol 500mg	Painkiller	ReliefMax	50
Aspirin 75mg	Painkiller	CardioGuard	70
Dextromethorphan 30mg	Cough Suppressant	CoughAway	80
Simvastatin 20mg	Cholesterol	LipiSafe	50
Salbutamol 2mg	Respiratory	AirEase	30
Diazepam 5mg	Anxiety	NeuroCalm	50
Erythromycin 250mg	Antibiotic	Bactrocin	45
Naproxen 500mg	Painkiller	FlexaRelief	40
Clindamycin 300mg	Antibiotic	ClindaCare	40
Carvedilol 12.5mg	Cardiovascular	HeartSure	30
Allopurinol 300mg	Gout Treatment	UricAid	60
Hydrocortisone 20mg	Corticosteroid	DermaCort	40

Low Stock Report

Low Stock Report

PharmacyInventory Edited 5 days ago by you

This question is written in SQL.

drug_name	category	brand_name	quantity_in_stock
Ferlin	Iron Supplement	HemoIron	10
Trulicity	Diabetes	GlucoTune	10
Penmenvy	Antiviral	MediCure	10
Morphine	Painkiller	MS Contin	12
Neo-Bromexan Forte	Cold Medication	FluBlock	12
Resmetirom	Liver Disease	HepatoCare	12
Octanate	Hemophilia	FactorX	12
Xultophy	Diabetes	GlucoBlend	13
Trastuzumab	Cancer	OncoShield	13
Glarsulin	Diabetes	InsuFlow	14
Benadryl	Antihistamine	HistaRelief	14
Albuterol	Asthma	BreatheWell	14

Expiration Monitoring Report

Expiration Monitoring Report

PharmacyInventory Edited 5 days ago by you

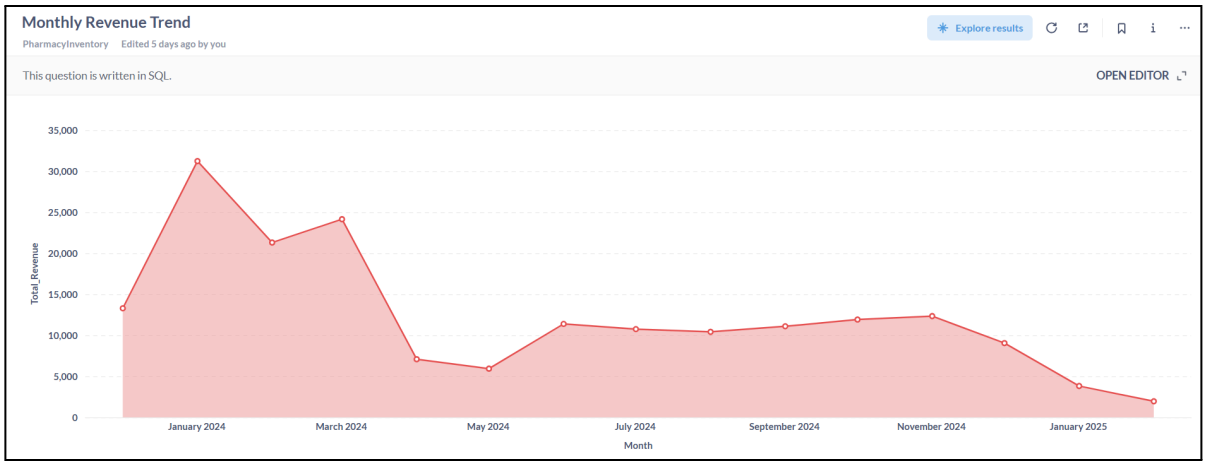
This question is written in SQL.

drug_name	category	brand_name	expiration_date
Oseltamivir	Antiviral	Tamiflu	April 5, 2025
Biofil	Wound Healing	DermaHeal	April 6, 2025
Immubulin	Immunoglobulin	BactoShield	April 8, 2025
Methotrexate	Cancer	ChemoRelief	April 12, 2025
Cenprahep	Hepatitis Treatment	HepatoShield	April 14, 2025
Clopidogrel	Antiplatelet	ThromboSafe	April 16, 2025
Buscopan	Antispasmodic	SpasmGuard	April 17, 2025
Teramol	Painkiller	PainMed	April 17, 2025
Metrophenicol	Antibiotic	BactoStop	April 20, 2025
Jusline N	Diabetes	SugarReg	April 21, 2025
Duspatalin	Antispasmodic	SpasmEase	April 21, 2025
Beratcant	Respiratory Medic	LungShield	April 21, 2025

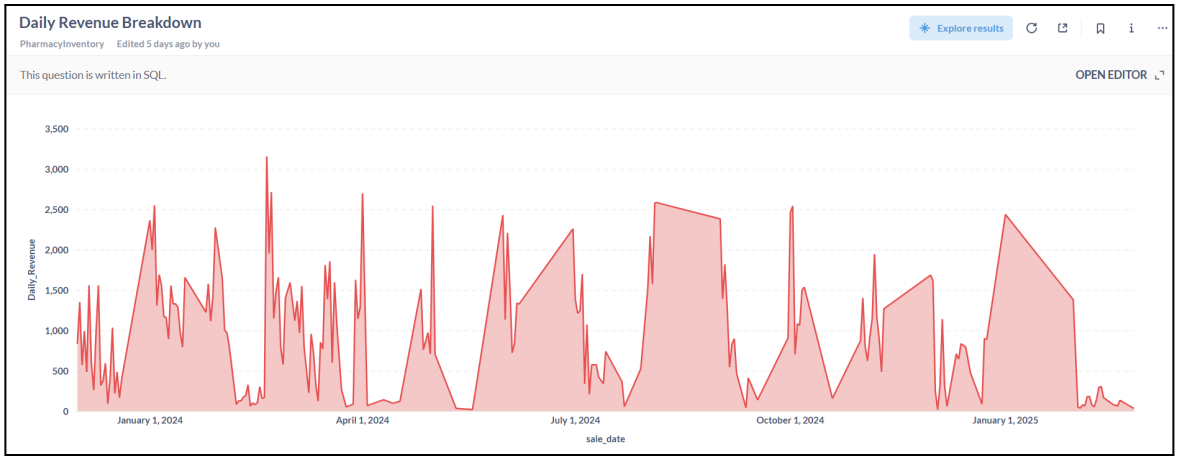
Dropdown #2: Sales & Revenue Performance

Sales Dashboard:

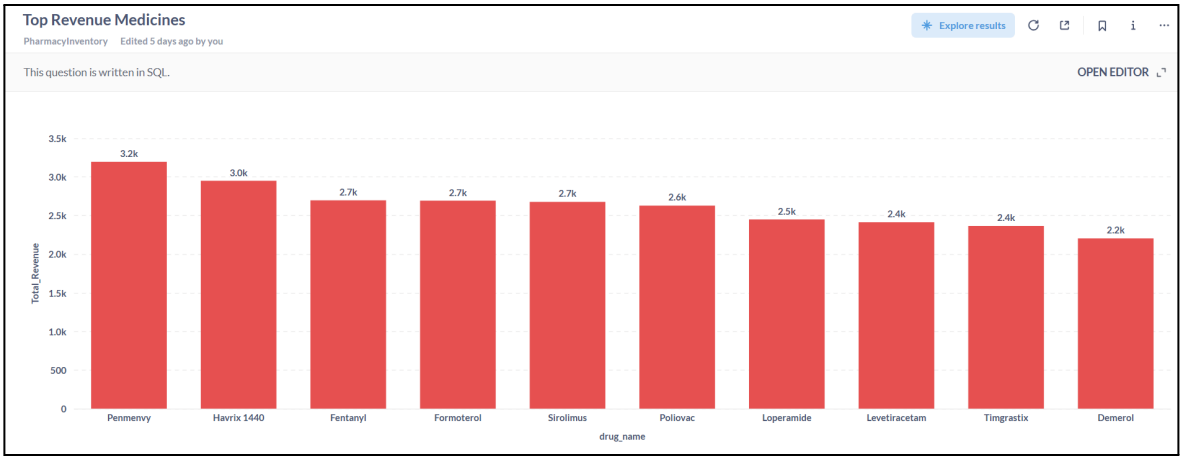
Monthly Revenue Trend



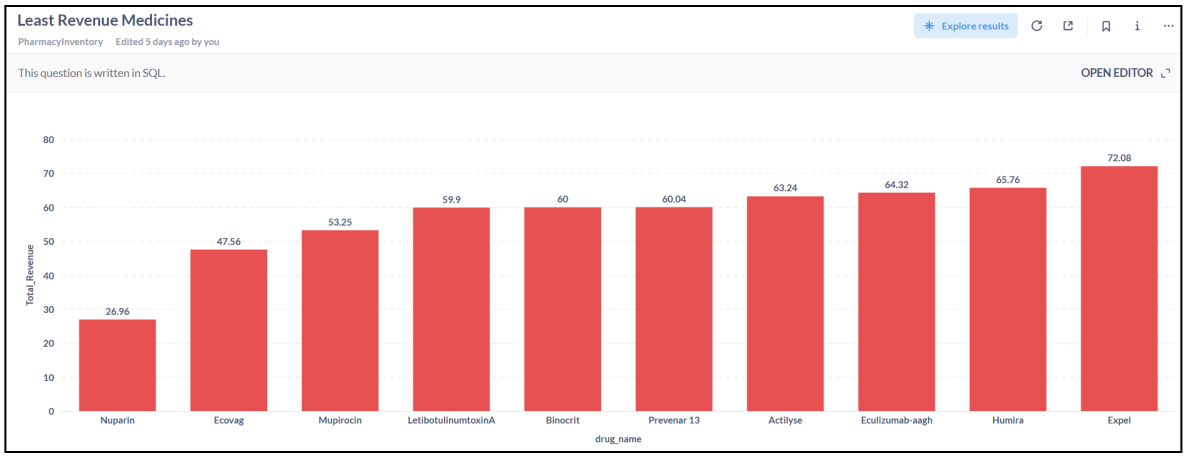
Daily Revenue Breakdown



Top Revenue Medicines



Least Revenue Medicines



Daily Revenue Report

Daily Revenue Report

PharmacyInventory Edited 5 days ago by you

This question is written in SQL.

sale_date	Daily_Revenue
February 25, 2025	32.8
February 19, 2025	134.75
February 18, 2025	65.5
February 16, 2025	82
February 12, 2025	168.1
February 11, 2025	304.4
February 10, 2025	296.9
February 9, 2025	145.6
February 8, 2025	55.5
February 7, 2025	74.5
February 6, 2025	180.8
February 5, 2025	181.2

Revenue by Medicine

Revenue by Medicine Report

PharmacyInventory Edited 5 days ago by you

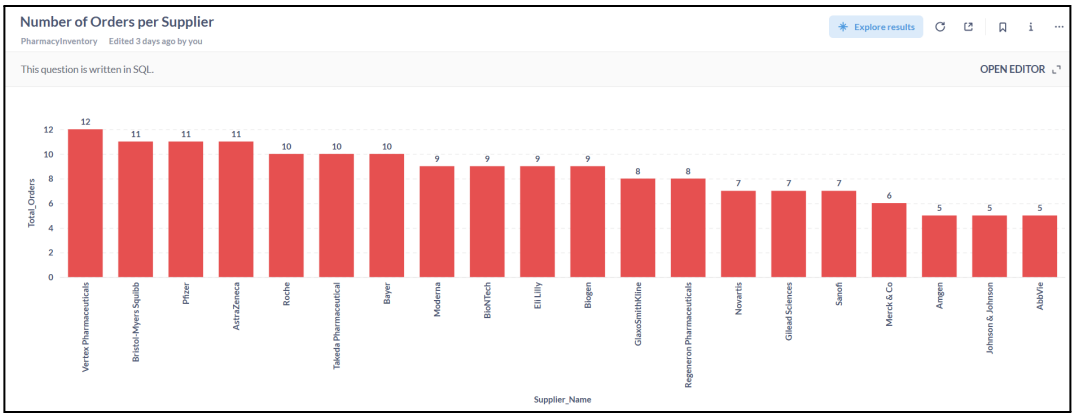
This question is written in SQL.

drug_name	Total_Revenue
Penmenvy	3,196.67
Havrix 1440	2,949.96
Fentanyl	2,697.3
Formoterol	2,692.8
Sirolimus	2,677.89
Poliovac	2,629.86
Loperamide	2,450
Levetiracetam	2,413.83
Timgrastix	2,366.07
Demerol	2,204.73
Pulmoxcel	2,149.48
Fluticasone	2,091.5

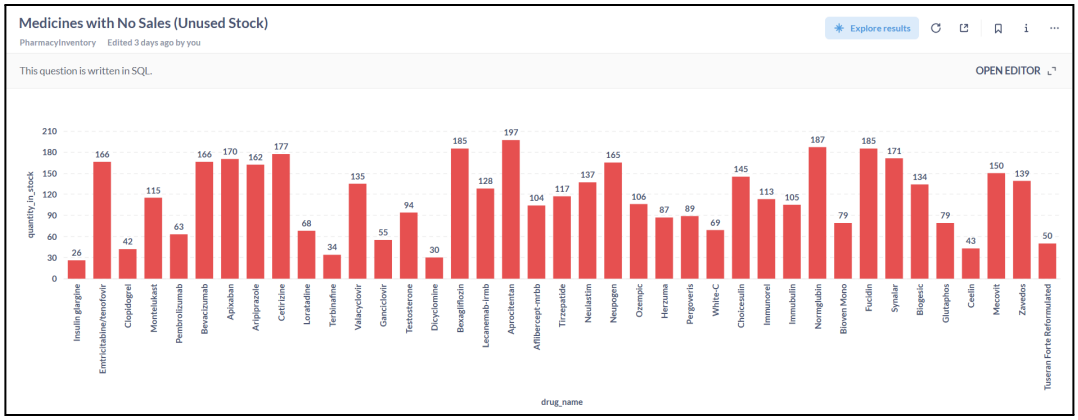
Dropdown #3: Supplier & Procurement Overview

Supplier & Procurement Dashboard:

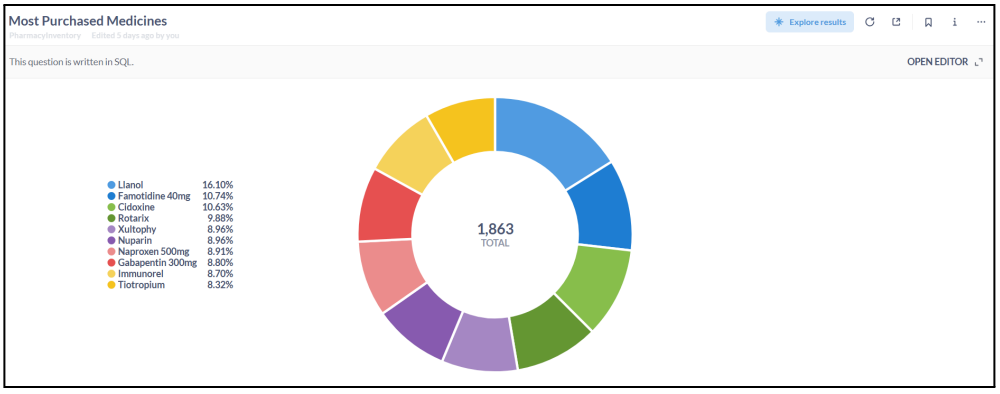
Number of Orders per Supplier



Medicines with No Sales (Unused Stock)



Most Purchased Medicines



Controlled Medicines Prescribed (Last 30 Days)

Controlled Medicines Prescribed (Last 30 Days)

PharmacyInventory Edited 5 days ago by you

This question is written in SQL.

doctor_name	patient_name	Medicine_Name	date_prescribed
Dr. Carter	Ace	Glutaphos	April 4, 2025
Dr. McCoy	Amy	Dolutegravir	September 7, 2025
Dr. Gatwa	Susan	Fucidin	November 15, 2025
Dr. Jones	Clara	Glysolin G	June 1, 2025
Dr. McGann	Rory	Cyclophosphamide	March 25, 2025
Dr. Bishop	Jack	Ozempic	June 6, 2025
Dr. Palmer	River	Betaloc	November 22, 2025
Dr. Harper	Clara	Nitroglycerin	October 20, 2025
Dr. Morgan	Jo	Ganciclovir	June 9, 2025
Dr. Sullivan	Sarah	Ledipasvir/sofosbuvir	September 30, 2025
Dr. Harper	Clara	Pulmoxcel	March 24, 2025
Dr. Davidson	Yasmin	Expel	May 12, 2025

Supplier Orders Summary

Supplier Orders Summary

PharmacyInventory Edited 5 days ago by you

This question is written in SQL.

Supplier_Name	Total_Orders
Vertex Pharmaceuticals	12
Bristol-Myers Squibb	11
Pfizer	11
AstraZeneca	11
Roche	10
Takeda Pharmaceutical	10
Bayer	10
Moderna	9
BioNTech	9
Eli Lilly	9
Biogen	9
GlaxoSmithKline	8

Unused Medicines Report

Unused Medicines Report

PharmacyInventory Edited 5 days ago by you

This question is written in SQL.

drug_name	quantity_in_stock
Insulin glargine	26
Emtricitabine/tenofovir	166
Clopidogrel	42
Montelukast	115
Pembrolizumab	63
Bevacizumab	166
Apixaban	170
Aripiprazole	162
Cetirizine	177
Loratadine	68
Terbinafine	34
Valacyclovir	135

Controlled Medicines Report

Controlled Medicines Report

PharmacyInventory Edited 5 days ago by you

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doctor_name	patient_name	Medicine_Name	date_prescribed
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Dr. Harper	Clara	Pulmoxcel	March 24, 2025
Dr. Davidson	Yasmin	Eymal	May 12, 2025

USER GUIDE

To use the EMC001 Dashboard:

1. ***Launch the Web Application***
    a. Open index.html in your browser.
    b. The dashboard loads with the sidebar on the left and topbar navigation.
2. ***Toggle Sidebar or Theme***
    a. Click the menu icon to collapse or expand the sidebar.
    b. Click the sun/moon icon to switch between light and dark mode.
3. ***Navigate to Reports***
    a. Select a section (e.g., Medicine & Inventory) and click on a specific report.
    b. The page loads a dashboard or report using an embedded Metabase iFrame.
4. ***View Reports***
    a. Use the iFrames to interact with charts, filters, or hover to view data points.
5. ***Download Reports***
    a. Click the "Download PDF Report" button (for sections that support it).